Maren

CONFIDENTIAL

Maren Finance
Investment & Grant Thesis

Mezo's Bitcoin credit primitive works. The constraint is distribution. Maren is the consumer layer that removes it.

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1

Executive summary

Mezo's core primitive works. A Bitcoin holder can post BTC, borrow MUSD, and never sell. That is the hard part, and it is already shipped, live on mainnet, and correct.

The constraint on Mezo is not the primitive. It is that the primitive has almost no consumer use. This is what the chain actually shows, read from mainnet:

MEASURE	VALUE	READ FROM
MUSD total supply	~31 million	MUSD token contract
MUSD created by user borrowing	~4.6% of that	`TroveManager.getEntireSystemDebt()`
Sitting in the Stability Pool (farm)	~46% of supply	Stability Pool balance
Held by the top 20 addresses	~99.5%	token holders, almost all contracts
Troves open on Mezo mainnet	~27	`TroveManager.getTroveOwnersCount()`

Re-run `evidence/VERIFY.sh` before quoting any of these; the counts drift. What does not drift is the shape. MUSD today is a DeFi-internal asset: minted mostly by the protocol, parked in yield strategies and AMM pools, and barely borrowed by users at all. The four largest troves hold roughly 80% of all borrowed MUSD. There is no consumer economy for MUSD. Nobody spends it, because there is nowhere to spend it and no accessible way for a normal-sized holder to create it.

We want to be precise about what that gap is and is not. It is **not** evidence that tens of thousands of people are trying to borrow and cannot; the holder base is a long tail of dust, and the real MUSD economy is about twenty contracts farming yield. It **is** evidence of two real, fixable frictions: the protocol has a **2,000 MUSD minimum loan** (about $3,000 of Bitcoin to open responsibly, $5,000 to be safe), which prices out small holders entirely; and once you hold MUSD there is no consumer surface to use it on.

Maren Finance is a bet that a consumer layer creates the demand the protocol cannot reach on its own. A pooled vault with **no minimum loan size** (deposit from 0.001 BTC, borrow what you want), gasless so a user never holds the asset they refused to sell, and payments to an `@handle` so MUSD finally has somewhere to go. This is demand creation, stated as a bet, not a captured market. The evidence for it is off-chain, in Mezo's large community, not yet on-chain in the trove count.

The economics are unusually favourable here. Mezo's base fee is 1,300,000 wei, putting a 100,000-gas transaction at about **$0.008**: under a cent at any BTC price below roughly $77,000. On most chains gasless is a subsidy with an end date. On Mezo it is a structural property that can be held indefinitely.

Maren is not a first attempt. It is the second, by a builder who shipped a payments system on another chain, commissioned an audit of it, and published the result, seven Critical and thirteen High severity findings, in public, with the remediations.

The ask is the ecosystem grant of up to $15,000 MUSD, 40% released on mainnet deployment and the balance against the milestones in section 8. Spent entirely on sponsored gas it would cover approximately 1.8 million user transactions; it is sized by what an audit costs, not by what gasless costs.

Section 9 prices the risks: pooled-trove systemic risk, a relayer hot key, an upgradeable MUSD proxy, oracle staleness, and pre-audit status.

2

The opportunity

Mezo has done the difficult, unglamorous work. MUSD is a Threshold-USD / Liquity-v1 fork with real risk parameters, a live liquidation engine, an oracle fed by the validator set, and a Stability Pool that pays out in BTC. The collateral is native BTC passed as `msg.value`: no wrapper, no bridge, no synthetic. A Bitcoin holder can draw dollars against Bitcoin and keep the Bitcoin. That is the whole promise of the chain and it functions today.

The parameters are conservative and, importantly for a consumer product, *stable*:

PARAMETER	VALUE	NOTES
Minimum collateral ratio	`110%`	Hardcoded `constant`, not governable
Critical (recovery) ratio	`150%`	Hardcoded `constant`, not governable
Borrowing fee	`0.10%`	Governable, 7-day timelock
Interest rate	`1.00% APR`	Simple, non-compounding, fixed for the life of the loan
Redemption fee	`0.75%`	Taken in BTC, paid to the PCV
Governance delay	`7 days`	On every governable parameter

A fixed 1.00% simple APR that a trove keeps forever is a genuinely good consumer credit product. It is cheaper than every card, most mortgages, and every other on-chain Bitcoin-backed loan we are aware of. It does not float, it does not compound, and a global rate change does not touch a loan already open.

So the product is good. The distribution is the problem, and the distribution problem is not vague. It is three mechanical frictions, each with a number attached.

3

The three frictions

3.1

The minimum is too big

`BorrowerOperations.minNetDebt()` returns 1,800 MUSD. Opening a trove also escrows a 200 MUSD gas compensation, refunded on close, so the minimum total debt at open is **2,000 MUSD**.

What that costs in Bitcoin depends on how much liquidation risk a user accepts. At the liquidation edge (110%) it is about **$2,200**; at a responsible ratio it is **$3,000 to $5,000**. Call it a $3,000 door to open a small position without sitting on the cliff. That is smaller than the $5,004 we used to quote (that was the safe-250% figure), and we have corrected it everywhere.

The real point is not that the collateral is expensive. It is that **the loan has a minimum size at all**: you cannot borrow 50 MUSD against Bitcoin on Mezo, only 2,000 or more. Maren's pooled vault removes that minimum entirely, which is a different and stronger claim than "cheaper collateral". These figures come from `collateralRequiredFor()` in `packages/sdk/src/position.ts` against the live oracle, so anyone who clones the repository can check them.

> NOTE
>
> This is not a criticism of the parameter. A 1,800 MUSD floor is the correct choice for a protocol that has to make liquidations economic: below it, the gas compensation and the liquidator's 0.5% share stop covering the cost of liquidating. The floor is right for the protocol and wrong for the consumer. That is exactly the kind of gap an application layer exists to close.

The consequence is visible on-chain. Mezo mainnet carries about **27 troves**, and four whales hold roughly 80% of all the MUSD ever borrowed. Small borrowers do exist, at the edge: the smallest live trove is about 0.03 BTC running at a 116% collateral ratio, one move from liquidation. That is the whole population the minimum loan pushes to the cliff.

We are deliberately not claiming this proves pent-up borrowing demand. It does not. It proves the minimum makes the safe version of a small position impossible, and that the few who try anyway run dangerously. Whether enough people want to borrow small against Bitcoin to spend is the bet Maren is making, and the honest place that bet rests is Mezo's off-chain community, not its trove count.

3.2

Every action costs BTC gas

Mezo's gas token is BTC. That is coherent and it is part of the chain's identity. It also produces a specific absurdity at the consumer layer: a user borrows MUSD precisely so they do not have to sell Bitcoin, and then must hold a Bitcoin balance in order to move the MUSD.

For a $5,000 borrower this is a rounding error. For the user Maren is built for, someone with 0.01 BTC who wants $200 of spending power, being told to first acquire and hold a separate BTC balance for gas is a hard stop. It is also a conceptual stop: the product's promise is "you do not have to touch your Bitcoin", and the first screen asks them to.

The fix costs almost nothing on this chain, which is the point of section 4.

3.3

There is nowhere consumer-grade to spend it

MUSD exists. A payments surface for it does not. Sending MUSD today means a hex address, a manually-managed BTC balance for gas, and no memo, no request, no split, no link, no receipt. That is a settlement primitive, not a payments product.

Nobody adopts a dollar because it exists. They adopt it because their rent, their contractor, and their friend can all receive it in one action.

4

The wedge

Maren addresses all three frictions in application code. None of it requires Mezo to change a parameter, upgrade a contract, or grant a permission.

FRICTION	TODAY ON MEZO	WITH MAREN
Minimum to borrow	0.079346 BTC ($5,004.50)	**0.001 BTC** ($63.07)
BTC needed to transact	Yes, for every action	**None**
Send MUSD to	A hex address	`@handle`, link, request, or split

4.1

Pooled Bitcoin credit, from 0.001 BTC

`MarenVault` runs one large, conservatively managed trove and gives each depositor a proportional claim inside it, with per-user collateral and debt accounting and a per-user ICR computed off the same oracle the protocol uses.

The floor is **0.001 BTC**, and more to the point there is no minimum loan at all: you can borrow 25 MUSD, which the protocol alone does not permit at any collateral level. Maren enforces an internal liquidation ratio of **140%** against the protocol's 110%, so the pooled trove is never operated near the protocol cliff, and a keeper tops up collateral, repays debt, and internally liquidates unhealthy users before the protocol would ever see the position. Users above the protocol minimum are steered to a **direct** trove instead, which carries no pooling risk at all, and the interface says so plainly.

This is the differentiator, and it is also where Maren's risk lives. Section 9 does not soften it.

4.2

Gasless, at a cost that does not bite

Mezo's base fee is **1,300,000 wei**. A 100,000-gas transaction therefore costs 1.3×10^{-7} BTC, **about $0.008** at BTC $63,000, and under one cent at any BTC price below roughly $77,000.

SPONSORED ACTIONS	APPROXIMATE COST
1,000	$8
100,000	$820
1,000,000	$8,200

VERIFIED

Derived from the live base fee of 1,300,000 wei and BTC at $63,072 as reported by Mezo's own oracle precompile on 2026-08-15. At a planning price of $70,000 the per-transaction figure is $0.0091 and a million actions cost $9,100. The conclusion is insensitive to the price: gasless on Mezo is cheap at any Bitcoin price a Bitcoin holder would be happy about.

Mechanically, Maren combines an ERC-2771 trusted forwarder with MUSD's EIP-2612 `permit`, so a first-time user can approve *and* transact in a single signed action while holding zero BTC. For direct troves we use Mezo's own `BorrowerOperationsSignatures` rather than wrapping it, fewer moving parts and no additional trust assumption.

This matters strategically, not just operationally. On most chains, sponsored transactions are a growth subsidy with an end date; the unit economics force the training wheels off. On Mezo the subsidy is a rounding error against any plausible revenue line, so it never has to stop. Gasless can be a permanent property of the product rather than a campaign.

4.3

Payments against `@handles`

`MarenRegistry` maps `@sele` to an address, with commit-reveal registration to stop front-running and reverse resolution for display. `MarenPay` operates on MUSD and covers direct payment with an on-chain memo, payment requests, claimable links, and one-signature splits to many recipients. Nothing is custodial, MUSD moves wallet to wallet under the user's own permit.

A claimable link is the specific unlock. The recipient needs no prior relationship with Maren, no MUSD, and no BTC. That is the mechanism by which an existing Mezo user brings someone who has never heard of Mezo onto the chain.

5

Why this is good for Mezo specifically

This is not a general-purpose wallet that happens to support Mezo. Every part of the argument is chain-specific.

Every Maren user is a new MUSD borrower. Maren does not compete for the existing troves. It addresses the population underneath the 1,800 MUSD floor, which is currently zero and which is where the ~29,000 MUSD holders mostly live. The pooled vault mints new MUSD against new BTC collateral; the direct-trove path sends larger users to Mezo's own contracts unchanged.

Every Maren action is a Mezo transaction. Payments, handle registrations, vault adjustments, and Earn deposits all settle on Mezo and pay Mezo's fees, and because Maren sponsors the gas, those fees get paid whether or not the user holds BTC.

Maren's own contracts deploy on Mezo. `MarenVault`, `MarenPay`, `MarenRelayHub`, `MarenRegistry`, and `MarenEarn` are Maren-authored Solidity written, tested, and bound for 31612. The activity, the fees, and the TVL accrue to the chain, not to a chain Maren likes better.

Maren builds with the grain of what Mezo already shipped. Mezo operates its own gasless payment rail, the x402 settlement proxy at `0x402085c248EeA27D92E8b30b2C58ed07f9E20001`, live on both networks, where MUSD's permit support lets a buyer authorise with a single EIP-712 signature and no approval transaction. Maren interoperates with that rail rather than building a competing one. Where Mezo ships a signed-operation contract, Maren calls it instead of wrapping it.

5.1

What Mezo gets beyond the application

Building Maren has required reading Mezo's deployed bytecode rather than its documentation, and that produced findings worth handing back regardless of the grant:

FINDING	STATUS
viem's bundled `mezo` chain definition uses an RPC host that fails DNS, and omits `multicall3` although Multicall3 is deployed	Corrected definition shipped in `packages/sdk/src/chains.ts`; upstreamable
Three hosts published in Mezo's own docs return NXDOMAIN: `rpc.mezo.org`, `testnet-rpc.mezo.org`, `testnet-explorer.mezo.org`	Confirmed; a correcting pull request is prepared
Three places where Mezo's published docs contradict the deployed contracts	Enumerated in `packages/sdk/src/constants.ts`, offered to Mezo as findings
ERC-4337 is deployed on Mezo but structurally unusable, and the reason is specific	Written up in the feasibility document

These are offered as contributions, not complaints. A chain with a thin tooling layer benefits from builders who write down what they had to discover.

6

Traction and credibility

6.1

What is already built, and how you can check

The contracts are not a proposal. Everything in this table is in the repository today and reproducible from a clean checkout, which is the only kind of traction a pre-launch project can honestly offer.

COMPONENT	STATE	HOW TO CHECK IT
Core contracts	Five, all under the 24,576-byte limit, `MarenVault` is 14,508 bytes	`forge build --sizes`
Contract test suite	**236 tests passing**, 8 of them fork tests against real matsnet MUSD	`forge test`
Gasless relaying	`MarenRelayHub`, atomic signed-fee ERC-2771	`MarenRelayHub.sol`
Relayer service	**214 tests passing**, 6 against a live Mezo testnet anvil fork	`apps/relayer`
Landing page	17/17 behavioural checks, 8/8 contrast probes, 112 kB first load, fully static	`apps/web/scripts/verify-*`
The 0.079346 BTC boundary	Asserted by an executable test, not merely computed in prose	`MarenVault.t.sol`

The last row matters more than its size suggests. The number this entire document rests on, that a user needs $5,004.50 of Bitcoin before Mezo's CDP will speak to them, is now pinned by a test that fails if the protocol's own constants move. It decomposes as 1,800 MUSD net debt plus a 1.8 MUSD origination fee plus 200 MUSD gas compensation, taken to a 250% ratio at an oracle price of $63,071.83. If that figure is wrong, the build goes red.

6.2

The team

Maren is new. The person building it is not.

Talise is a payments application on Sui, built by the same author for Sui Overflow and carried past the hackathon into production. What it has, stated as measured:

MEASURE	VALUE
Waitlist signups	1,900+
Handles claimed on-chain	1,400+
Live mainnet deployments	Payments, cheques, streams, payroll, yield router, payment kit
Published security audit	2026-07-24

There is no TVL figure here, no revenue figure, and no monthly-active number, because those are not figures we can substantiate. A grant committee should assume any number absent from this table does not exist.

6.3

The part that should actually persuade you

Talise commissioned a security audit of its own codebase and **published it**. It found **7 Critical and 13 High severity issues**. The accompanying `SECURITY.md` labels parts of the system custodial, discloses that the shielded pool is not a trustless setup and has a small anonymity set, and documents that an unauthenticated command-injection route reached production for approximately six minutes on 2026-07-24, after which all in-scope secrets were rotated. It states, in the author's own words, that "a security document that only lists successes is not a security document."

Most teams applying for a grant present their first system. This application presents a second system, designed by someone holding a specific written list of what went wrong in the first one. Maren's architecture encodes those lessons directly:

- the relayer's funding source is bound to the caller, never inferred;
- nothing is keyed by a shared address, so no two users can ever collide in escrow;
- funding is verified on-chain rather than trusted from a client-asserted digest;
- access gates fail closed;
- the proxy admin is behind a timelock and a multisig from deployment day one;
- local authentication uses loopback with CSRF protection rather than custom URL schemes.

The same discipline is applied to Maren's own documentation. Talise's written docs describe an Enoki, dApp Kit, and Walrus Sites integration that never actually shipped, the generated codebase docs are the accurate map, and the prose was aspirational. That is precisely the failure mode found on Mezo's side too: documentation promising a borrower redemption-fee waiver that does not exist in the bytecode, and RPC hosts that do not resolve. Maren's documents are therefore written from verified deployed behaviour, and where something is intent rather than fact, it is labelled as intent.

7

Business model

Maren has no token. Talise has no token and no tokenomics, and Maren will not launch one. In a Bitcoin-native ecosystem, “we are not printing a token” is a feature, and it removes the single largest source of misaligned incentives from the design.

Revenue comes from spread on real usage. The natural lines on Mezo:

LINE	BASIS
Pooled vault spread	A small margin over the protocol’s 1.00% APR, charged to pooled borrowers for the service of being under the minimum
Payments fee	A fee on MUSD payment volume above a free tier; the first 25 sponsored actions per address are always free
Earn performance fee	A performance fee on realised `MarenEarn` yield, never on principal

For scale, Talise’s model ranks four revenue lines, FX/auto-swap spread at 30 bps, off-ramp margin at 50 bps, yield rebate at roughly 100 bps of the lending spread, and on-ramp take at 50 bps, which produce an illustrative $14.10 per active user per year, or $1.41M at 100,000 users.

> WARNING
>
> Those unit economics are **illustrative, not achieved**. The constants exist in Talise’s code; they are not wired into a live revenue flow, and no revenue has been collected against them. They are included to show the shape of the model, not to imply a run rate. Maren’s equivalent lines are not live either. What is verifiable today is that at $0.008 per sponsored transaction, gas is a rounding error against any of them.

8

The ask

An ecosystem grant of up to $15,000 MUSD, with 40% ($6,000) released on mainnet deployment and the remaining $9,000 released against the milestones below, plus the co-marketing support the ecosystem team offers.

MILESTONE	RELEASE	DEFINITION OF DONE
M0, Mainnet deployment	$6,000	All five Maren contracts verified on 31612; pooled vault live with a cap; addresses published
M1, Audited	$3,000	Third-party audit of `MarenVault` and `MarenRelayHub` complete and published, findings and remediations in the open
M2, Borrowers under the floor	$3,000	100 distinct addresses holding a pooled position below the 1,800 MUSD protocol minimum, that is, 100 borrowers Mezo could not otherwise have
M3, Payments live	$3,000	1,000 settled `MarenPay` transactions and 500 registered handles, all gasless

The milestones are deliberately written as counts of things Mezo does not have today rather than as TVL. TVL can be bought. A hundred borrowers below a floor that currently admits nobody cannot.

How the money is used, in order: audit engagement first, relayer gas float second, infrastructure third. Development is not funded by this grant; it is already underway.

> NOTE
>
> On the gas float specifically: $15,000 spent entirely on sponsored transactions would cover approximately 1.8 million user actions at Mezo's current base fee. The grant is not sized by what gasless costs, it is sized by what an audit costs.

9

Risks, stated plainly

A committee that has to discover a risk after funding has been given a worse document than one that names it first.

Pooled-trove systemic risk. This is the largest one and it is inherent to the design. Users in `MarenVault` share a single trove. A user's collateral is segregated *in accounting*, not in custody. A BTC drawdown faster than the keeper can respond could socialise losses across pooled depositors. Mitigations, the 140%-versus-110% buffer, per-block keeper checks, the 0.001 BTC deposit floor, halting new borrowing before halting withdrawals, and an insurance buffer funded from vault fees, reduce the probability. They are not a guarantee, and the product interface says exactly that. Users large enough to open a direct trove are steered to one, which has none of this risk.

Relayer key risk. The relayer holds a hot key funded with BTC. It is allowlisted to Maren contracts and known selectors only, rate-limited per address and per IP, and it never holds user funds. A compromise costs Maren its gas float. It cannot move a user's MUSD or collateral, because every user action is authorised by the user's own EIP-712 signature and MUSD moves under the user's own permit.

MUSD is upgradeable. Every core MUSD contract is a `TransparentUpgradeableProxy` controlled by Mezo's multisig. Maren integrates against the proxies, as one must, which means Maren inherits whatever the proxy admin does. Governable parameters carry a 7-day timelock; the implementation behind the proxy is a trust assumption in Mezo, and it belongs on this list even though it is not Maren's to fix.

Oracle risk. The BTC/USD feed MUSD consumes is Skip Connect, Mezo's own consensus-level oracle module, exposed as a chain precompile at `0x7b7c000000000000000000000000000000000015`. It is not Chainlink and it is not Pyth: it is secured by Mezo's validator set rather than by an independent third party. It reverts if the price is more than 60 seconds stale, which means every borrow, adjust, and liquidate path can revert for reasons that have nothing to do with the user, and a liquidation cannot execute while the feed is down. Both Maren clients have an explicit "price feed unavailable" state; we never render a stale price as if it were live.

Pre-audit. Maren is pre-audit today. `MarenVault` and `MarenRelayHub` are the two contracts that carry risk and are the audit scope. Mainnet deployment ships with the vault capped, and the cap is raised only as the position proves out. The first use of grant money is the audit engagement.

Execution risk. Maren is a small team building a consumer application on a chain with a thin tooling layer, no public subgraph, and a short list of working RPC endpoints. The feasibility document quantifies exactly what that costs and what the fallbacks are, which is the honest way to state it.

10

What we are asking you to fund

Not a new primitive. Mezo already built the primitive and it works.

We are asking you to fund the layer that takes a $5,004.50 minimum down to $63.07, removes the requirement to hold Bitcoin in order to spend against Bitcoin, and gives MUSD somewhere to go, on a chain where the gas to do all of that costs less than a cent a transaction.

Mezo's borrower count is in the twenties because of a floor, not because of a lack of interest. There are more than 29,000 people already holding the stablecoin. Maren is the shortest path from that number to the other one.

The technical case, including the contract-level integration surface, the seven ways MUSD diverges from Liquity, and the reasons the obvious account-abstraction path is a dead end on this chain, is set out in **MRN-002, Technical Feasibility & Architecture**.

Every protocol parameter, address, and chain fact in this document was read live from Mezo RPCs on 2026-08-15 and is recorded in machine-readable form in `packages/sdk/src/constants.ts` and `packages/sdk/src/addresses.ts`. Where Mezo's published documentation contradicts the deployed bytecode, this document follows the bytecode and says so.

Figures describing Talise are stated as measured. Figures describing Maren's future are stated as projections and labelled as such. Nothing in this document is an audited claim; Maren is pre-audit and says so in section 9.